

RiboFUSION: Supplementary Material

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This supplementary material supports the audit claims of the main text. The unfavourable results are reported where they arise in the main text; here we collect the complete per-method data and the provenance note so that every headline claim can be re-read against one frozen source.

S1. Frozen per-method metrics. Tables S1, S2, and S3 list every per-method metric for the three locked cohorts, read from a single frozen results table; no value is recomputed after locking. The RiboFUSION aggregate and its USDF-5 aggregate coincide on the internal cohort by construction of the locked value.

Table S1: Frozen per-method metrics, internal cohort (bold = RiboFUSION aggregate).

Method	AUC	AP	Brier	LogLoss	P@1	R@20	ECE
RiboFUSION	0.8663	0.7261	0.249	1.1287	0.9045	0.5485	0.259
USDF5	0.8663	0.7261	0.249	1.1287	0.9045	0.5485	0.259
USDF4	0.8722	0.7372	0.2262	1.8967	0.9361	0.5534	0.2254
RiboEncoder	0.8707	0.7263	0.147	0.4591	0.9182	0.5457	0.1179
MHCFlurry	0.8383	0.6734	0.1994	0.5941	0.8864	0.5152	0.2281
BigMHC	0.8379	0.6884	0.1416	0.5006	0.9045	0.53	0.0831
TransPHLA	0.8253	0.6283	0.2939	0.9621	0.7682	0.4958	0.3278
PLM-LR	0.7917	0.5724	0.1579	0.4866	0.8	0.4476	0.0395

Table S2: Frozen per-method metrics, Ouspenskaia cohort (bold = RiboFUSION aggregate). A dash marks a metric not computed for that method.

Method	AUC	AP	Brier	LogLoss	P@1	R@20	ECE
RiboFUSION	0.9207	0.9057	0.1529	1.7702	1.0	0.3794	0.1497
USDF5	0.914	0.9001	0.1629	0.4988	1.0	0.3776	0.1544
USDF4	0.9095	0.9006	0.1604	0.4948	1.0	0.378	0.1473
RiboEncoder	0.8451	0.8428	0.1608	0.4878	0.9935	0.3632	0.0175
MHCFlurry	0.8837	–	0.216	0.6337	0.987	–	–
BigMHC	0.9265	–	0.2082	–	0.9805	–	–
TransPHLA	0.8991	0.8648	0.1937	0.6565	0.9091	0.3564	0.2136
PLM-LR	0.8077	0.7781	0.1795	0.5433	0.8961	0.3337	0.0343

S2. Unfavourable results, tabulated. Each unfavourable result is stated in place in the main text; Table S4 collects the values so that the negatives can be read together without reopening the results.

S3. Provenance and code. Every number in the paper is read from one frozen results table; the three provenance values in Fig. 4N are anchored against a 20k recompute, and the USDF-4

Table S3: Frozen per-method metrics, Chong cohort (bold = RiboFUSION aggregate).

Method	AUC	AP	Brier	LogLoss	P@1	R@20	ECE
RiboFUSION	0.9481	0.918	0.1131	1.6165	1.0	0.3589	0.087
USDF5	0.9438	0.9359	0.14	0.4376	0.875	0.3883	0.1714
USDF4	0.9422	0.9319	0.1339	0.4239	1.0	0.3883	0.1541
RiboEncoder	0.9043	0.8916	0.1312	0.4237	0.875	0.3792	0.082
MHCFlurry	0.9199	0.9065	0.1234	0.3983	0.875	0.3815	0.0963
BigMHC	0.9336	0.9304	0.1203	0.3893	1.0	0.3928	0.0899
TransPHLA	0.9238	0.899	0.1894	0.6324	1.0	0.3657	0.22
PLM-LR	0.8481	0.8178	0.1593	0.4891	0.875	0.3431	0.0498

Table S4: The unfavourable results disclosed in the main text, with the section in which each is reported.

Unfavourable result	Value	Reported in
Per-allele AUROC spread (rare bucket)	0.55–0.93	Sec. 4.4
Weak IG-to-edit correlation	0.229	Sec. 5
Negative step of the N3 reader	−0.0040	Sec. 4.6
Internal component reversal	0.8707 > 0.8575 > 0.8663	Sec. 4.7
P@1 four-way tie (Chong)	0.875	Sec. 4.3

Ousp value is the live-recomputed 0.9095 (a stale 0.9096 present in an earlier official file is not cited). The figure-generation scripts, the protocol lock, and the per-figure data files are released with the repository named in the data-availability statement below.